

Exercise 2.7

Given

$$F_0(x) = 1 - \frac{1}{1+x}, \quad x \geq 0,$$

find expressions for (a), (b), (c) below, simplifying as far as possible,

(a) $S_0(x)$,

(b) $f_0(x)$,

(c) $S_x(t)$,

and calculate:

(d) p_{20} , and

(e) ${}_{10|5}q_{30}$.

Solution

(a)

The survival function is

$$S_0(x) = 1 - F_0(x).$$

Thus,

$$\begin{aligned} S_0(x) &= 1 - \left(1 - \frac{1}{1+x}\right) \\ &= \frac{1}{1+x}. \end{aligned}$$

Therefore,

$$\boxed{S_0(x) = \frac{1}{1+x}}.$$

(b)

The density function is

$$f_0(x) = \frac{d}{dx} F_0(x).$$

Since

$$F_0(x) = 1 - (1 + x)^{-1},$$

we have

$$\begin{aligned} f_0(x) &= \frac{d}{dx} [1 - (1 + x)^{-1}] \\ &= (1 + x)^{-2} \\ &= \frac{1}{(1 + x)^2}. \end{aligned}$$

Thus,

$$\boxed{f_0(x) = \frac{1}{(1 + x)^2}.$$

(c)

For a life aged x , the survival function for a further t years is

$$S_x(t) = \frac{S_0(x + t)}{S_0(x)}.$$

Using $S_0(x) = 1/(1 + x)$,

$$\begin{aligned} S_x(t) &= \frac{\frac{1}{1 + x + t}}{\frac{1}{1 + x}} \\ &= \frac{1 + x}{1 + x + t}. \end{aligned}$$

Therefore,

$$\boxed{S_x(t) = \frac{1 + x}{1 + x + t}.$$

(d)

Since p_{20} denotes the one-year survival probability for a life aged 20,

$$p_{20} = S_{20}(1).$$

Using the formula from part (c),

$$\begin{aligned} p_{20} &= \frac{1 + 20}{1 + 20 + 1} \\ &= \frac{21}{22} \\ &= 0.954545. \end{aligned}$$

Thus,

$$\boxed{p_{20} = \frac{21}{22} \approx 0.954545}.$$

(e)

The probability ${}_{10|5}q_{30}$ means surviving 10 years from age 30 and then dying within the next 5 years. Hence,

$${}_{10|5}q_{30} = {}_{10}p_{30} {}_5q_{40}.$$

Now,

$${}_{10}p_{30} = S_{30}(10) = \frac{1 + 30}{1 + 30 + 10} = \frac{31}{41}.$$

Also,

$${}_5p_{40} = S_{40}(5) = \frac{1 + 40}{1 + 40 + 5} = \frac{41}{46},$$

so

$${}_5q_{40} = 1 - {}_5p_{40} = 1 - \frac{41}{46} = \frac{5}{46}.$$

Therefore,

$$\begin{aligned} {}_{10|5}q_{30} &= \frac{31}{41} \left(1 - \frac{41}{46} \right) \\ &= \frac{31}{41} \cdot \frac{5}{46} \\ &= 0.082184. \end{aligned}$$

Thus,

$$\boxed{{}_{10|5}q_{30} \approx 0.082184}.$$